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Stranded seaweed –

an environmentally friendly insulation material?

According to latest research, six out of nine planetary boundaries are already crossed indicating the risk of irreversible environmental changes (Stockholm Resilience Centre, 2022). One of the main drivers responsible for these adverse environmental impacts is the buildings and construction sector causing about 37% of carbon dioxide emissions due to energy demand in the production and use phase, 100 billion tonnes of waste and a significant amount of resource consumption (United Nations Environment Programme, 2022). Bio-based insulation materials provide a solution to reduce energy demand for heating and cooling (United Nations Environment Programme, 2022) while promising a low environmental impact (Geß et al., 2021). In the following it is going to be assessed if this promise holds true for seaweed insulation material. It can be produced from *Posidonia oceanica*, *Laminaria digitata*, or *Zostera* species whose leaves accumulate at the shore as waste material that is landfilled (Gräbe, 2013; Kuqo & Mai, 2022). Due to its insulating properties seaweed can instead be reused as insulation material (Benjeddou et al., 2022; Bouasria et al., 2021; Brandhorst et al., 2012; Geß et al., 2021; Holzmann & Wangelin, 2009a). But what is the environmental impact of the use of stranded seaweed as loose insulation material?

First, the seaweed grows naturally and just needs to be collected by hand at the shore, thus "the production of the raw material is considered burden free" according to Geß et al. (2021, p. 7) who conducted a life cycle analysis for seaweed insulation. More specifically, the seaweed material has a low acidification and eutrophication potential as it does not require agricultural production (Geß et al., 2021). It also does not require additional land which is a huge advantage in comparison to other natural insulation materials (Benjeddou et al., 2022). Instead, the reuse of stranded seaweed prevents landfilling. On top of that, the seaweed provides various ecosystem services such as preventing soil erosion at the coast (Bouasria et al., 2021; Brandhorst et al., 2012; Holzmann & Wangelin, 2009c) and draws down carbon dioxide from the atmosphere through photosynthesis leading to a negative global warming potential for raw material production (Geß et al., 2021). Further processing of the seaweed only involves cleaning, sieving and cutting (Greiner et al., 2011). Overall, the production process is relatively short, requires little energy (Gräbe, 2013; Kuqo & Mai, 2022) and creates only few emissions (Geß et al., 2021). In the use phase the seaweed can substitute emission-intensive petroleum-based materials (Benjeddou et al., 2022) or wood-based insulation materials which need energy-intensive refining and can have negative impacts due to deforestation (Kuqo & Mai, 2022; Widera, 2014). Additionally, the insulation leads to energy savings (Geß et al., 2021). Therefore, out of the four sustainability strategies listed by Holzmann and Wangelin (2009b), the seaweed insulation addresses the substitution and efficiency strategy. Regarding disposal, the seaweed can either be composted or incinerated which creates carbon dioxide emissions and therefore has a big impact on the global warming potential of the seaweed insulation (Geß et al., 2021; Kuqo & Mai, 2022). However, the incineration can be used for energy generation, thus substituting a part of Germany's grid mix (including energy from fossil fuels) (Geß et al., 2021). Geß et al. (2021) subtracted the amount of emissions that can be avoided due to this substitution from their life cycle analysis, thus arriving at negative global warming, ozone depletion and abiotic depletion potentials. This approach is questionable as the negative values represent avoided additional pollution (relative to Germany's current unsustainable grid mix) rather than an actual drawdown of pollutants from the atmosphere.

Overall, seaweed had the lowest emissions of all studied renewable and conventional insulation materials (Geß et al., 2021). This holds true even if the credits for grid mix substitution are taken out of the calculation. The outcome is, however, limited to seaweed harvested naturally as large seaweed monocultures cause damage for example to benthic ecosystems (Titlyanov & Titlyanova, 2010). Furthermore, the longer the transport the more is emitted which limits the use of seaweed to a local scale (Widera, 2014). It must also be considered that seaweed meadows are currently declining due to climate change, invasive species and human activities, so that the supply in the future is uncertain (Telesca et al., 2015). In conclusion, the environmental impact of seaweed insulation can be very low, but only if it is used on a small-scale regional basis. Because of that and the uncertain supply in the future it can be one of multiple environmentally friendly insulation materials contributing to a diversification of the construction sector instead of a panacea. To assess the sustainability and desirability of seaweed insulation, however, economic and social implications need to be considered, too.

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